INCH-POUND

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COMMERCIAL ITEM DESCRIPTION

CUTTER BIT BLANKS AND CUT-OFF BLADES

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers cutter bit blanks and cut-off blades of high speed steel that are supported by tool holders and used to perform various metal-working machining operations.

2. CLASSIFICATION. The cutter bit blanks and cut-off blades shall be of the following types, styles, classes, and sizes.

2.1 Type. The type of cutter bit blanks and cut-off blades shall be as specified (see 7.3(b)).

Type I - Cutter bit blank
Type II - Cut-off blade

2.2 Style. The style of cutter bit blanks and cut-off blades shall be as specified (see 7.3(c)).

Style A - Square cross-section (see table I)
Style B - Rectangular cross-section (see table II)
Style C - Straight side clearance, one end beveled (see table III)
Style D - Straight side clearance, both ends beveled (see table III)
Style E - Armstrong style (see table IV)

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Supply Center Richmond (DSCR), ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616.

AMSC N/A FSC 3455

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2.3 Class. The class of cutter bit blanks and cut-off blades shall be as specified (see 7.3(d)).

Class 1 - M-1 steel industry designation (general purpose applications)
Class 2 - M-2 steel industry designation (general purpose applications)
Class 3 - M-33 steel industry designation (heavy duty applications)
Class 4 - M-34 steel industry designation (heavy duty applications)
Class 5 - M-43 steel industry designation (heavy duty applications)
Class 6 - M-46 steel industry designation (heavy duty applications)
Class 7 - T-15 steel industry designation (heavy duty applications)

2.4 Size. The size and dimensions of the cutter-bit blanks and cut-off blades shall conform to table I, II, III, and IV (see 7.3(e)). The tolerance of all dimensions shall be the manufacturer's standard as specified on their drawings.

TABLE I. Type I, style A, square cross-section cutter bit blanks.

Size	Square cross-section (inches)	Length (inches)
01	1/8	2-1/2
02	3/16	2-1/2
03	1/4	2-1/2
04	5/16	2-1/2
05	3/8	3
06	7/16	3-1/2
07	1/2	4
08	9/16	4
09	5/8	4-1/2
10	3/4	5
11	7/8	6
12	1	7
13	1-1/8	8
14	1-1/4	9
15	1-1/2	10

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TABLE II. Type I, style B, rectangular cross-section cutter bit blanks.

Size	Dimensions		
	Width (inches)	Height (inches)	Length (inches)
16	1/4	3/8	2-1/2
17	1/8	1/2	3
18	1/4	3/8	3
19	3/8	1/2	3
20	5/16	1/2	3
21	5/16	7/16	3-1/2
22	1/8	1/2	4
23	1/4	1/2	4
24	3/8	1/2	4
25	3/8	5/8	4
26	3/8	3/4	4
27	1/2	3/4	4
28	3/8	5/8	4-1/2
29	1/4	3/4	5
30	3/8	5/8	5
31	3/8	3/4	5
32	1/2	3/4	5
33	5/8	3/4	5
34	1/4	1/2	6
35	1/4	1	6
36	3/8	1/2	6
37	3/8	5/8	6
38	3/8	3/4	6
39	3/8	1	6
40	1/2	3/4	6
41	1/2	1	6
42	5/8	7/8	6
43	3/4	1	6
44	1/2	1	7
45	3/4	1	7
46	1/2	1	8

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TABLE III. Type II, styles C and D, straight side clearance cut-off blades.

Size	Dimensions		
	Thickness (inches)	Height (inches)	Length (inches)
47	5/64	1/2	4
48	1/16	1/2	4-1/2
49	3/32	1/2	4-1/2
50	1/16	11/16	5
51	3/32	5/8	5
52	3/32	11/16	5
53	1/8	11/16	5
54	1/16	13/16	6
55	3/32	13/16	6
56	1/8	13/16	6
57	5/32	13/16	6
58	3/16	13/16	6
59	1/4	13/16	6
60	3/32	1	6-1/2
61	1/8	1	6-1/2
62	3/16	1	6-1/2
63	1/4	1	6-1/2
64	5/16	1	6-1/2

TABLE IV. Type II, style E, Armstrong style cut-off blades.

Size	Dimensions		
	Thickness (inches)	Height (inches)	Length (inches)
65	3/32	1/2	4-1/2
66	3/32	5/8	5
67	1/8	3/4	5
68	1/8	7/8	6
69	3/16	1	6-1/2
70	3/16	1-1/8	6
71	1/4	1-1/4	7
72	1/4	1-1/8	7

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3. SALIENT CHARACTERISTICS

3.1 General requirements.

3.1.1 Materials. The cutter bit blank and cut-off bladed high speed tool steel shall conform to ASTM A 600, "Standard Specification for Tool Steel High Speed".

3.1.2 Hardness. The cutter bit blank and cut-off blade high-speed steel alloy shall be hardened and tempered to a Rockwell "C" hardness of not less than 63 when tested in accordance with ASTM E 18, "Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials".

3.2 Type I, cutter bit blanks.

3.2.1 Style A. Type I, style A cutter bit blanks shall meet the dimensional requirements listed in table I and shall have a square cross section with the ends beveled at an angle of 10 degrees.

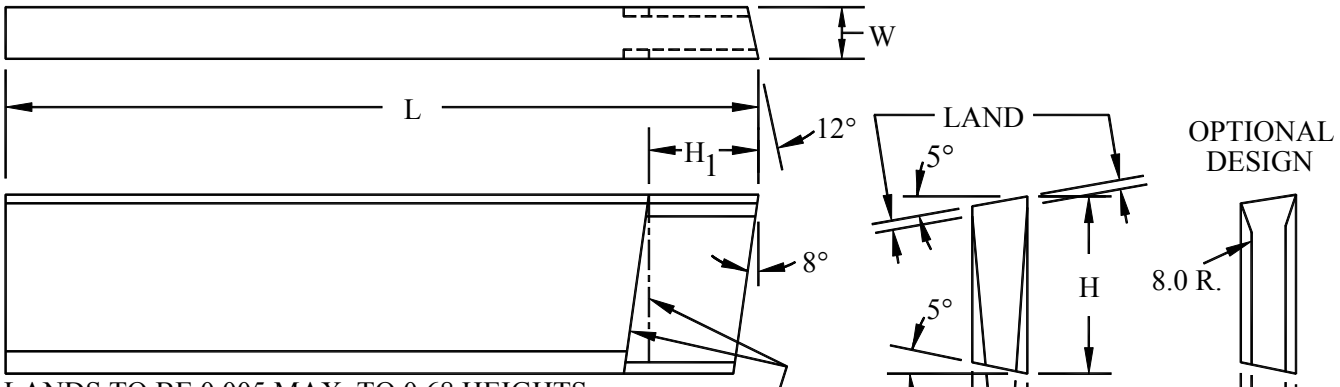
3.2.2 Style B. Type I, style B cutter bit blanks shall meet the dimensional requirements listed in table II and shall have a rectangular cross section with the ends squared.

3.3 Type II, cut-off blades.

3.3.1 Styles C and D. Type II, styles C and D cut-off blades shall meet the dimensional requirements listed in table III.

3.3.2 Style E. Type II, style E cut-off blades shall meet the dimensional requirements listed in table IV.

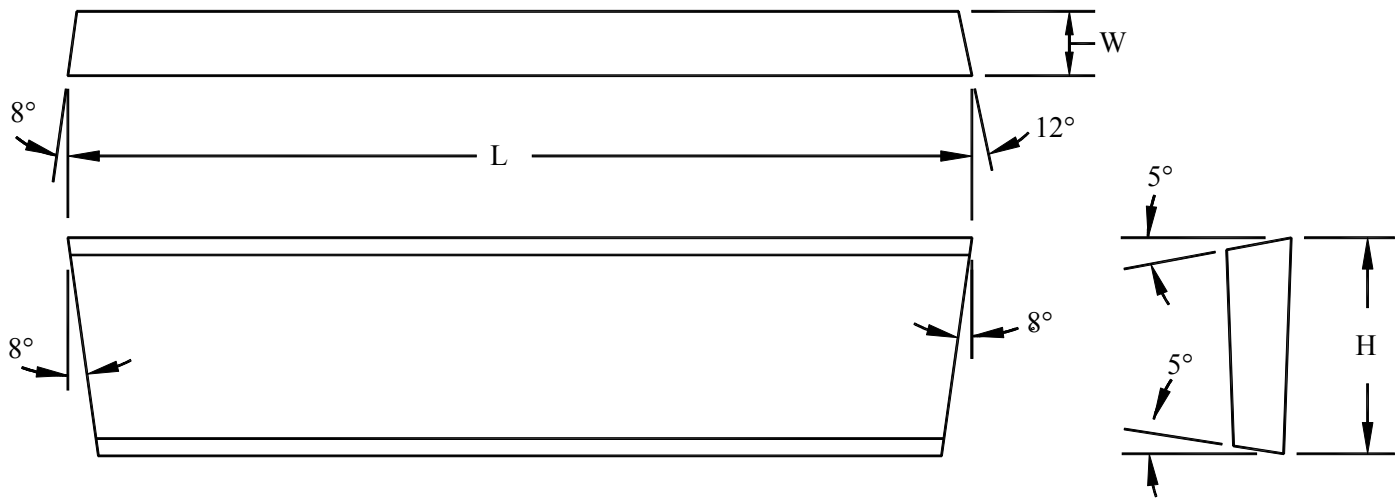
3.3.3 Cut-off blade designs. Styles C, D, and E cut-off blades shall conform to figures 1, 2, and 3 respectively.



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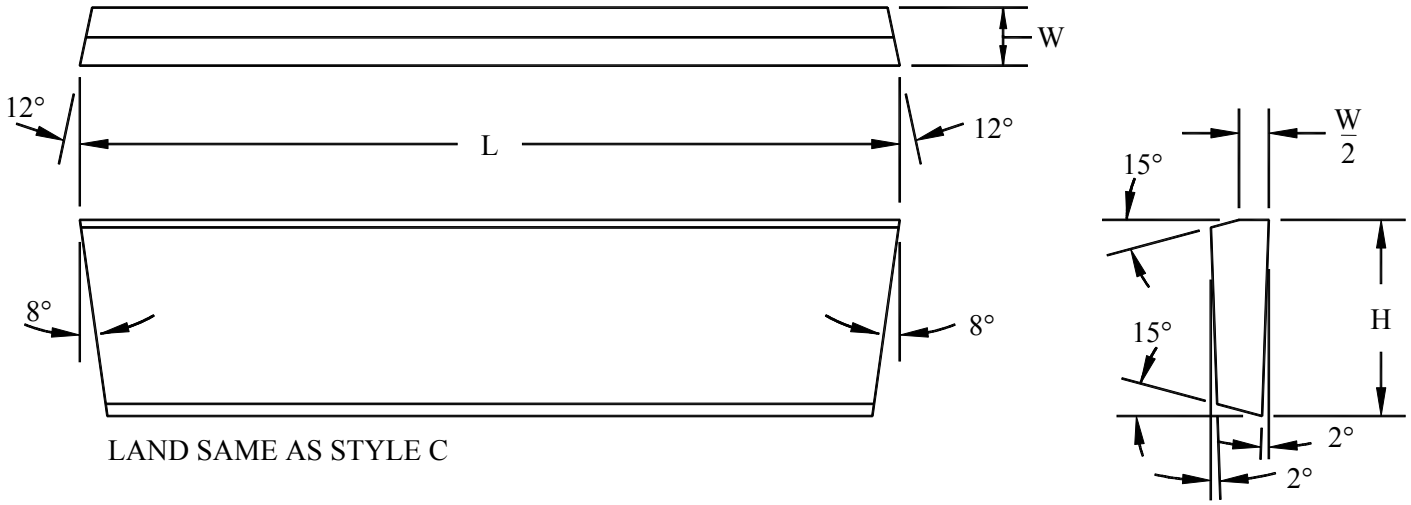
FIGURE 1. Style "C" cutoff blade.

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FIGURE 2. Style "D" cutoff blade.



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FIGURE 3. Style "E" cutoff blade.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

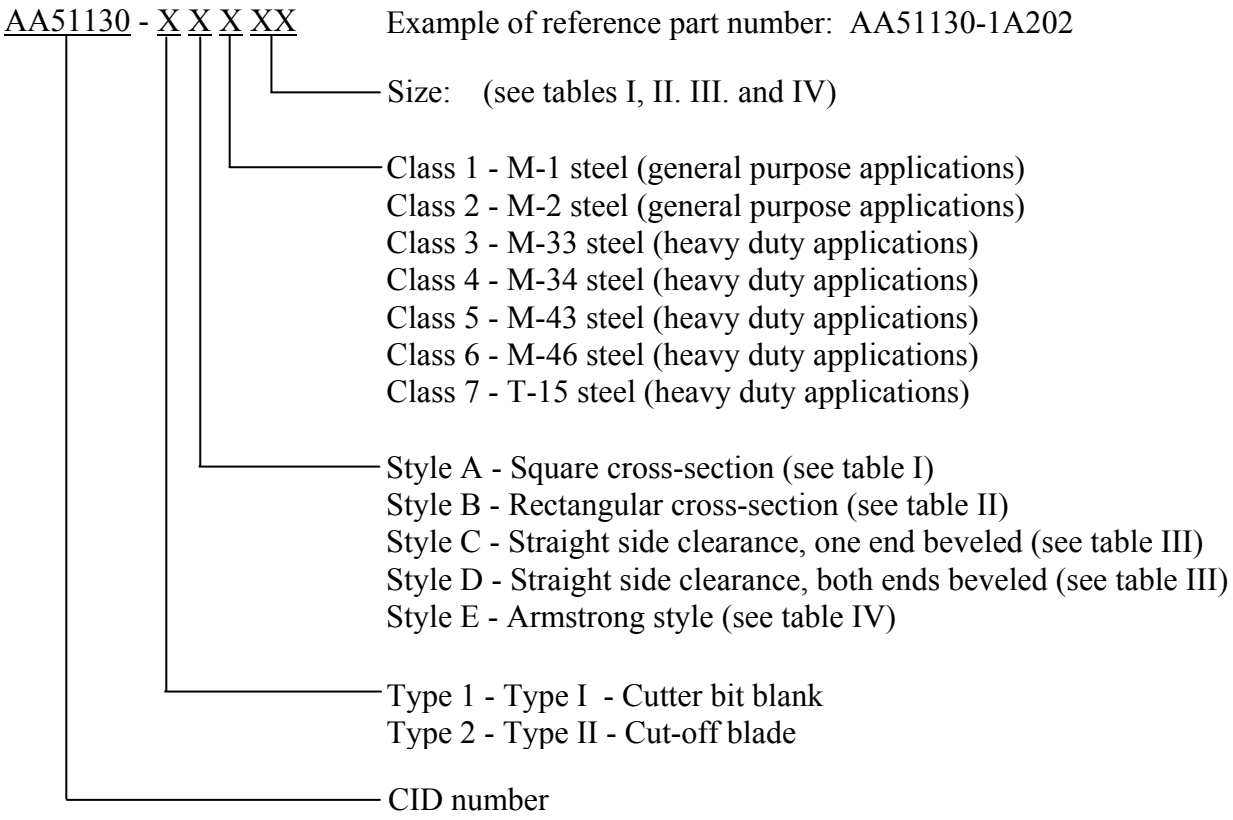
5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the acquisition order (see 7.3(f)).

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



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7.2 Sources of documents.

7.2.1 FAR. The FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of FAR documents may be obtained from http://www.arnet.gov/far/.

7.2.2 ASTM standards. Copies of ASTM standards may be obtained from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959. Electronic copies of ASTM standards may be obtained from http://www.astm.org/.

7.3 Ordering data. The acquisition order should specify the following:

a. CID document number, revision, and CID PIN.
b. Type of tool (see 2.1).
c. Style of tool (see 2.2).
d. Class of tool (see 2.3).
e. Size of tool (see tables I, II, III, or IV).
f. Preservation, packing, and marking requirements (see 6.1).

7.4 Subject term (key word) listing.

machining
metal working

MILITARY INTERESTS:	CIVIL AGENCY COORDINATING ACTIVITY:
Custodian: Air Force - 99	GSA - FSS
Review Activity: Air Force - 84	Preparing Activity: DLA - GS6
	(Project 3455-2007-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at http://assist.daps.dla.mil.

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